

Natural Growth and Decay

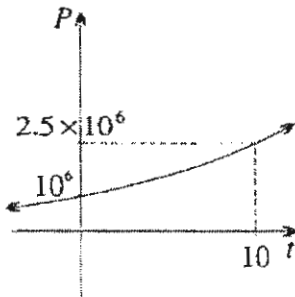
- Given that $y = 5e^{\frac{1}{2}t}$, show that $\frac{dy}{dt} = \frac{1}{2}y$. Use these equations to solve the following.
 - Find y to three significant figures when:
 - $t = 2$
 - $t = -3$
 - $t = 4.5$
 - Find t to two decimal places when:
 - $y = 10$
 - $y = 1$
 - $y = 30$
 - Find the exact value of $\frac{dy}{dt}$ when:
 - $y = 8$
 - $y = 11$
 - $y = \frac{1}{3}$
 - Find $\frac{dy}{dt}$ to three decimal places when:
 - $t = 7$
 - $t = 3.194$
 - $t = \ln 3$
- In the following questions, give exact answers where appropriate or else approximate to four significant figures.
 - Given that $y = 3e^{2t}$, show that $\frac{dy}{dt} = 2y$. Find y when $t = 1.5$.
 - Given that $y = e^{-t}$, show that $\frac{dy}{dt} = -y$. Find t when $y = \frac{1}{2}$.
 - Given that $y = 10e^{-\frac{1}{2}t}$, show that $\frac{dy}{dt} = -\frac{1}{2}y$. Find $\frac{dy}{dt}$ when $y = 6$.
 - Given that $y = \sqrt{2}e^{\sqrt{2}t}$, show that $\frac{dy}{dt} = \sqrt{2}y$. Find $\frac{dy}{dt}$ when $t = \frac{1}{\sqrt{2}}$.
- The population P of a city rose from 1 million at the beginning of 1975 to 2.5 million at the beginning of 1985. Assuming natural growth, $P = 10^6 \times e^{kt}$ where t is the time in years since the beginning of 1975.
 - Find the value of the positive constant k , and sketch the curve.
 - What was the population of city at the beginning of 1998, to three significant figures?
 - In what year is the population 10 million?
 - Find the rate $\frac{dP}{dt}$ at which the population is increasing at the beginning of that year. Give your answer to the nearest thousand.
- Ten kilograms of sugar is gradually dissolved in a vat of water. After t hours, the amount of undissolved sugar remaining is given by $S = 10e^{-kt}$.
 - Calculate k , given that $S = 3.2$ when $t = 4$, and sketch the graph.
 - At what time will there be 1 kg of sugar remaining?
 - How fast is the sugar dissolving after 1 hour? Give your answer in units of kilograms per hour to the nearest half kilogram.
- The value V of a factory machine depreciated with time t years such that $\frac{dV}{dt} = -kV$, for some constant $k > 0$.
 - Show that $V = V_0e^{-kt}$ satisfies the given differential equation.
 - The initial value of a particular item of machinery is \$15000. Explain why $V_0 = 15000$.
 - In the first year the machine depreciates in value by 30%. Find the value of the constant k .
 - The company that bought the machine writes off any machine when it has depreciated to 5% of its initial value. How many years does this take? Round your answer up to the nearest whole year.
- In an experiment in which bacteria are grown on a petri dish, it is found that the area A in cm^2 covered by the bacteria increases from 0.5 cm^2 to 1 cm^2 in a period of 3 hours.
 - Assuming that the area covered obeys the law of natural growth, show that A at time t hours after the initial observation is given by $A = \frac{1}{2}e^{kt}$.
 - Find the value of the positive constant k .
 - What area of the petri dish will be covered after 7 hours? Answer to three significant figures.
 - If the diameter of the petri dish is 10 cm, how long will it take for the bacteria to cover the dish? Answer to the nearest 10 minutes.

7. When a liquid is placed in a refrigerator kept at 0°C , the rate at which it cools is proportional to its temperature h at time t , thus $\frac{dh}{dt} = -kh$ where k is a positive constant.
- Show that $h = h_0 e^{-kt}$ is a solution of the differential equation.
 - Find h_0 , given that the liquid is initially at 100°C .
 - After 5 minutes the temperature has dropped to 40°C . Find the value of k .
 - Find the temperature of the liquid after 15 minutes.
8. The height H of a wave decays away so that $H = H_0 e^{-\frac{1}{3}t}$, where H_0 is the initial height of the wave. Giving your answer to the nearest whole percent, what percentage of the initial height is the height of the wave when:
- $t = 1$
 - $t = 3$
 - $t = 8$
9. A quantity Q of radium at time t years is given by $Q = Q_0 e^{-kt}$, where k is a positive constant and Q_0 is the amount of radium at time $t = 0$.
- Given that $Q = \frac{1}{2} Q_0$ when $t = 1690$ years, calculate k .
 - After how many years does only 20% of the initial amount of radium remain? Give your answer to the nearest year.
10. The most efficient way of boiling water is to add heat at a rate proportional to the temperature H of the water, thus $\frac{dH}{dt} = kH$.
- Show $H = H_0 e^{kt}$ is a solution of the differential equation.
 - The water initially at room temperature 20°C , and after $1\frac{1}{2}$ minutes reaches 30°C . Find the value of k .
 - Find how long, to the nearest second, it takes for the water to reach boiling point.
11. The amount A in grams of carbon-14 isotope in a dead tree trunk after t years is given by $A = A_0 e^{-kt}$, where A_0 and k are positive constants.
- Show that A satisfies the equation $\frac{dA}{dt} = -kA$.
 - The amount of isotope is halved every 5750 years. Find the value of k .
 - For a certain dead tree trunk the amount of isotope is only 15% of the original amount in the living tree. How long ago did the tree die (answer to the nearest 1000 years)?
12. Current research into Alzheimer's disease suggests that the rate of loss of percentage brain function is proportional to the percentage brain function already lost. That is, if L is the percentage brain function lost, then $\frac{dL}{dt} = kL$, for some constant $k > 0$.
- Two years ago a patient was initially diagnosed with Alzheimer's disease, with a 15% loss of brain function. This year the patient was diagnosed with 20% loss of brain function. Show that $L = 15e^{kt}$, where $k = \frac{1}{2} \ln \frac{4}{3}$.
 - The nearby care centre will admit patients to 24-hour nursing care when a patient reaches 60% loss of brain function. In how many more years will that be? Answer to the nearest year.
13. A wet porous substance loses moisture at a rate that is proportional to the moisture content M , that is $\frac{dM}{dt} = -kM$, where $k > 0$. On a particular day a wet towel on a clothes-line loses half its moisture in the first $1\frac{1}{2}$ hours.
- Show that $M = M_0 e^{-kt}$ is a solution of the differential equation.
 - Find the value of k .
 - How long, in total, will it take for the towel to become 99% dry? Answer to the nearest hour.
14. Air pressure P in millibars is a function of the altitude a in metres, with $\frac{dP}{da} = -\mu P$. The pressure at sea level is 1013.25 millibars.
- Show that $P = 1013.25 e^{-\mu a}$ is a solution to this problem.
 - One reference book quotes the pressure at 1500 metres to be 845.6 millibars. Find the value of μ for the data in that book.
 - Another reference book quotes the pressure at 6000 metres to be half that at sea level. Find the value of μ in this case.
 - Are the data in the two books consistent?
 - Assuming the first book to be correct:
 - What is the pressure at 4000 metres?

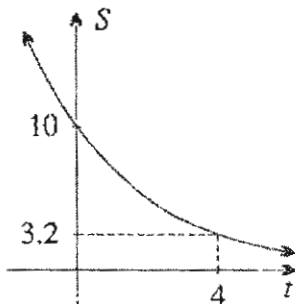
- ii. What is the pressure 1 *km* down a mine shaft?
 - iii. At what altitude is the pressure 100 millibars?
- 15. A certain radioactive isotope decays at such a rate that after 68 minutes only a quarter of the initial amount remains.
 - a. Find the half-life of this isotope.
 - b. What proportion of the initial amount will remain after 3 hours? Give your answer as a percentage to one decimal place.
- 16. The half-life of naturally occurring radioactive carbon C^{14} is 5730 years. In a living organism, the rate of C^{14} decay is typically constant at around 15.3 disintegrations per minute. When the organism dies the level of C^{14} decays away. Let C be the amount of radioactive carbon measured in disintegrations per minute at time t years.
 - a. Assuming natural decay, $C = C_0 e^{-kt}$. Find the values of k and C_0 .
 - b. In an archaeological dig, bones are found which exhibit 11 disintegrations per minute. How old are the bones, to the nearest year?
- c. Carbon dating is only deemed reliable for ages between 1000 and 10000 years. In another archaeological expedition, artefacts are found in a cave which exhibit 2.25 disintegrations per minute.
 - i. How old do these artefacts seem to be, to the nearest year?
 - ii. Is this figure reliable or should other tests be carried out to confirm the age to the artefacts?
- 17. In 1980 the population of Bedsworth was $B = 25000$ and the population of Yarra was $Y = 12500$. That year the mine was closed in Bedsworth and the population began falling, while the population of Yarra continued to grow, so that $B = 25000 \times e^{-pt}$ and $Y = 12500 \times e^{qt}$.
 - a. Ten years later it was found the populations of the two towns were $B = 20000$ and $Y = 15000$. Find the values of p and q .
 - b. In what year were the populations of the two towns equal?

Answer

- a. i. 13.6, ii. 1.12, iii. 47.4; b. i. 1.39, ii. -3.22, iii. 3.58;
c. i. 4, ii. $5\frac{1}{2}$, iii. $\frac{1}{6}$; d. i. 82.789, ii. 12.345, iii. 4.330
- a. $y \doteq 60.26$; b. $t \doteq 0.6931$; c. $\frac{dy}{dt} = -3$; d. $\frac{dy}{dt} = 2e$
- a. $k = \frac{1}{10} \ln \frac{5}{2} \doteq 0.092$; b. 8.23 million; c. during 2000;
d. $\frac{dP}{dt} = kP \doteq 916000$.



- a. $k = -\frac{1}{4} \ln \frac{3.2}{10} \doteq 0.28$; b. $t \doteq 8.08$ hours $\doteq 8$ hours 5 minutes; c. 2 kg/h



- b. V_0 is the value of V at $t = 0$; c. $-\ln \frac{7}{10} \doteq 0.36$; d. $\frac{1}{k} \ln 20 \doteq 9$ years
- b. $k = \frac{1}{3} \ln 2 \doteq 0.23$; c. 2.52 cm^2 ; d. 21 hours 50 minutes
- b. $h_0 = 100$; c. $k = -\frac{1}{5} \ln \frac{2}{5} \doteq 0.18$; d. 6.4°C
- a. 72%; b. 37%; c. 7%
- a. $k = \frac{\ln 2}{1690} \doteq 4.10 \times 10^{-4}$; b. $t = \frac{\ln 5}{k} \doteq 3924$ years
- b. $k = \frac{2}{3} \ln \frac{3}{2} \doteq 0.27$; c. 5 min 57 sec, to the nearest second
- b. $k = \frac{\ln 2}{5750} \doteq 1.21 \times 10^{-4}$; c. $t = \frac{1}{k} \ln \frac{100}{15} \doteq 16000$ years, to the nearest 1000 years
- b. 8 more years
- b. $k = \frac{2 \ln 2}{3} \doteq 0.46$; c. 10 hours
- b. $\mu_1 = 1.21 \times 10^{-4}$; c. $\mu_2 = 1.16 \times 10^{-4}$; d. The values of μ differ so the data are inconsistent; e. i. 625.5 millibars, ii. 1143.1 millibars, iii. 19205 metres
- a. 34 minutes; b. 2.5%
- a. $k = \frac{\ln 2}{5730} \doteq 1.21 \times 10^{-4}$, $C_0 = 15.3$; b. 2728 years old; c. i. 15847 years old, ii. Further tests should be carried out.
- a. $p = \frac{1}{10} \ln \frac{5}{4} \doteq 0.022$, $q = \frac{1}{10} \ln \frac{6}{5} \doteq 0.018$; b. $t = \frac{\ln 2}{p+q} \doteq 17.10$ years, that is during 1997.